

# Java Data Types

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# Data Types

## **Primitive Types:**

These types store simple values directly in memory and have high performance. These types are divided into numeric and non-numeric categories.

## **Reference Types:**

These types store the address (reference) of an object in memory. Their default value is null.

# Primitive Integer Types

## **byte**

- Size: 8 bits
- Range: -128 to 127

## **short**

- Size: 16 bits
- Range: -32,768 to 32,767

## **int**

- Size: 32 bits
- Range: -2,147,483,648 to 2,147,483,647

## **long**

- Size: 64 bits
- Range: -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807

# Primitive Floating-Point Types

## **float**

- Size: 32 bits
- Range: approx.  $1.4\text{e-}45$  to  $3.4\text{e+}38$
- Description: Single-precision floating-point

## **double**

- Size: 64 bits
- Range: approx.  $4.9\text{e-}324$  to  $1.8\text{e+}308$
- Description: Double-precision floating-point

# Primitive Non-Numerical Types

## **char**

- Size: 16 bits
- Range: '\u0000' to '\uffff' (0 to 65,535)
- Description: A single Unicode character

## **boolean**

- Size: JVM-dependent (typically 1 byte or 1 bit)
- Values: true or false
- Description: Logical value (truth value)

# Primitive Types Summary Table

| Type    | Size (bits)   | Range / Values                                          | Default Value |
|---------|---------------|---------------------------------------------------------|---------------|
| byte    | 8             | -128 to 127                                             | 0             |
| short   | 16            | -32,768 to 32,767                                       | 0             |
| int     | 32            | -2,147,483,648 to 2,147,483,647                         | 0             |
| long    | 64            | -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 | 0L            |
| float   | 32            | approx. 1.4e-45 to 3.4e+38                              | 0.0f          |
| double  | 64            | approx. 4.9e-324 to 1.8e+308                            | 0.0d          |
| char    | 16            | '\u0000' to '\uffff' (0 to 65,535)                      | '\u0000'      |
| boolean | JVM-dependent | true or false                                           | false         |

# Reference Types Categories

- **Class Types** – instances of classes
- **Interface Types** – references to objects that implement an interface
- **Array Types** – even arrays of primitives are objects
- **Enum Types** – a fixed set of named constants
- **Record Types** (Java 14+) – immutable data carriers with minimal boilerplate
- **Annotation Types** – used for metadata

# Wrapper Classes

Each primitive type has a corresponding wrapper class:

- `byte` → `Byte`
- `short` → `Short`,
- `int` → `Integer`
- `long` → `Long`
- `float` → `Float`
- `double` → `Double`
- `char` → `Character`
- `boolean` → `Boolean`

## Why use them?

- Required in generics
- Allow null as a value
- Provide utility methods